Department of Computer Science and Engineering
Academic Year: 2025 - 2026 (Odd Semester)

Name of the Faculty member	**: Mrs.S.Shunmuga Priya, AP/CSE**
Degree, Semester & Branch, Sec	**: B.E, V & CSE – 'B'**
Course Code & Title	**: CCS375 WEB TECHNOLOGIES**
Date & Time	**: 16.10.2025 & 12.40 PM to 1.05 PM**

Innovative Practice IV - Description

- Unit / Topic: Unit III / Servlet Life Cycle
- Course Outcome: CO3
- Topic Learning Outcome : TLO3a
- **Activity Chosen: Think – Pair - Share**

Justification

- This innovative practice was designed to strengthen students' conceptual understanding of the **Servlet Life Cycle** through an interactive and collaborative learning approach. In traditional teaching, servlet concepts are often delivered through lecture slides or code demonstrations, which can make the topic abstract and difficult for students to visualize.
- To address this, the **Think–Pair–Share** method was employed, allowing students to first think independently about how a servlet is created, serves requests, and is destroyed; then discuss with peers to clarify misconceptions; and finally share their understanding with the class through real-life analogies and live demonstrations.
- The objective of this Think–Pair–Share activity was to help students understand the Servlet Life Cycle and relate each phase to real-world systems, making the concept easier to visualize and remember through teamwork and live discussion.

Time Allotted for the Activity: 25 Minutes

Implementation:

The Servlet Life Cycle Think–Pair–Share activity was conducted in the following steps:

1. **Team Formation:**
   The class was divided into **3 teams**, each formulated a unique real-life system to correlate with the servlet life cycle:
   - **Team 1:** Chef (Restaurant)
   - **Team 2:** Teacher (Classroom)
   - **Team 3:** Amazon Online Shopping
2. **THINK:**
   Each student individually analyzed how a servlet goes through its life cycle stages (init(), service(), destroy()).
   Students then mapped these stages to the real-life system assigned to their team:
   - **Chef:** Kitchen preparation, serving orders, cleaning after service.
   - **Teacher:** Preparing lesson, teaching students, ending the class.

- **Amazon Shopping:** Loading product catalog, handling orders, closing sessions.

3. **PAIR:**
   Students paired within their teams to discuss their individual ideas, refine their analogies, and clarify misconceptions about the servlet life cycle stages as shown in Figure 1.
4. **SHARE:**
   Each team presented their understanding to the class as shown in Figure 2.
   - Team 1 (Chef): Demonstrated init() as preparing ingredients, `service()` as serving customers, `destroy()` as cleaning the kitchen.
   - Team 2 (Teacher): Explained init() as preparing lesson plans, `service()` as teaching students and answering questions, destroy() as ending the class and clearing resources.
   - Team 3 (Amazon Shopping): Showed init() as loading the system and product catalog, `service()` as processing customer orders, `destroy()` as closing sessions and cleaning temporary data.
5. After presentations, the instructor guided a discussion comparing the analogies and reinforcing the servlet life cycle concepts. Students reflected on how each real-life example maps to init(), service(), and destroy(), solidifying understanding through teamwork and visualization.

CO – PO / PSO mapping:

CO →	PO1	PO2	PO3	PO9	PO10	PO11	PSO1
CO3	3	3	3	1	2	2	3

PO-PSO Mapped:

Innovative Practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PO11**	**PSO1**
	3	3	3	1	2	2	3
Justification for Correlation	Students applied their engineering knowledge of web technologies to understand the servlet life cycle and its execution flow.	They analyzed and compared each servlet phase with real-life systems (Chef, Teacher, Amazon) to identify parallels and logic flow.	Students designed and demonstrated real-life analogies creatively, representing how servlets manage requests and responses.	The Think–Pair–Share method promoted teamwork and peer discussion, though each team mainly worked independently.	Students communicated effectively by explaining servlet processes and their real-life correlations to the class.	Teams planned and coordinated roles and presentations, reflecting organizational and management skills.	The activity enhanced practical understanding of Java web application development concepts through visualization and analogy.

- Images / Screenshot of the practice:

Figure 1

Figure 2

Reflective Critique
Learning Outcomes

- Conceptual Understanding: Students developed a clear understanding of the Servlet Life Cycle (init(), service(), destroy()) and its practical significance in web applications.
- Real-World Correlation: Students were able to relate abstract servlet phases to real-life systems like a chef in a restaurant, a teacher in a classroom, or Amazon's online shopping process.
- Active Learning & Engagement: The Think–Pair–Share method promoted active participation, peer discussion, and collaborative learning.
- Critical Thinking & Analysis: Students enhanced their ability to analyze processes and map technical concepts to familiar scenarios.
- Teamwork & Communication: Students improved team collaboration, presentation skills, and clarity in explaining technical topics.

Benefits of the Practice

- Enhanced Conceptual Clarity: Students understood abstract servlet concepts by relating them to real-life examples like a chef, teacher, or Amazon shopping system.
- Active Engagement: The Think–Pair–Share method encouraged active participation, discussion, and hands-on demonstration, making learning interactive.
- Improved Retention: Visualizing and enacting the servlet life cycle helped students remember concepts longer than traditional lectures.
- Development of Soft Skills: Students improved teamwork, communication, and presentation skills through collaborative activity.
- Bridging Theory and Practice: The activity linked theoretical knowledge with practical understanding, helping students grasp how web applications handle requests in real life.

Feedback of the Practice

- Students found relating servlet phases to real-life examples easy and understandable.
- The Think–Pair–Share method encouraged active participation and teamwork.
- The activity helped students visualize and retain concepts better than traditional lectures.

Challenges Faced in Implementation

- Some students initially struggled to map servlet phases to real-life examples.
- Differentiating between service() and request-specific methods like doGet()/doPost() required guidance.
- Time constraints limited the depth of each team's presentation and discussion.
- Coordinating team roles and ensuring equal participation was occasionally challenging.

References

https://www.readingrockets.org/classroom/classroom-strategies/think-pair-share
https://www.kent.edu/ctl/think-pair-share
https://ahaslides.com/blog/think-pair-share-activities/